Solubility and Colligative Properties

Molality, freezing and boiling points, osmotic pressure

Name _____ Class _____ Date _____

EQUATIONS AND DATA

$m = \text{mol solute} \div \text{kg solvent} \cdot \Delta T_f = i K_f m \cdot \Delta T_b = i K_b m \cdot \Pi = iMRT$

Water: $K_f = 1.86\ ^\circ\text{C kg/mol} \cdot K_b = 0.512\ ^\circ\text{C kg/mol} \cdot R = 0.08206\ \text{L atm mol}^{-1}\ \text{K}^{-1} \cdot T(\text{K}) = t(^\circ\text{C}) + 273.15 \cdot 1\ \text{atm} = 760\ \text{torr}$

Molar masses (g/mol): glucose 180.16 · $CaCl_2$ 110.98 · NaCl 58.44

Show units on every quantity in every step.

A. Understanding

1. Define (a) colligative property (b) molality (c) van ‘t Hoff factor. (d) Why does a solution boil at a higher temperature than the pure solvent?

2. Explain: (a) why sugar dissolves in water but not in hexane (b) why fish may struggle in warm water (c) why a fizzy drink bubbles when the cap is removed.

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B. Freezing and boiling points

3. 18.0 g of glucose is dissolved in 100. g of water. Find (a) the molality (b) the freezing point (c) the boiling point.

4. Put these 0.10 mol/kg aqueous solutions in order of freezing point, highest first: $CaCl_2$, glucose, NaCl. Explain.

5. 11.1 g of $CaCl_2$ is dissolved in 250. g of water. Find the freezing point (i = 3).

6. 3.00 g of a molecular compound dissolved in 50.0 g of water raises the boiling point by 0.256 °C. Find its molar mass.

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C. Osmotic pressure

7. 1.00 g of a protein is dissolved in water to make 100.0 mL of solution. Its osmotic pressure at 25.0 °C is 3.80 torr. Find the molar mass of the protein.

8. Blood has an osmotic pressure of about 7.70 atm at 37.0 °C. What concentration of NaCl (i = 2) would have the same osmotic pressure?

D. Challenge

9. What mass of NaCl must be dissolved in 1.00 kg of water to lower its freezing point to −10.0 °C (i = 2)? Suggest why calcium chloride is often used instead in very cold places.

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